EE3621: Digital Signal Processing

Lecture 13: IIR Filter Structures — Direct, Cascade & Parallel Forms
(III B.Tech EEE)

Lecture Notes (40-Minute Plan)

1 Lecture Plan (40 Minutes Breakdown)

- **00:00 – 05:00 (5 mins):** Welcome & Introduction to IIR Systems. Transfer function, difference equations, and flow graph basic elements.
- **05:00 – 12:00 (7 mins):** Direct Form I. Separate delay lines, node equations.
- **12:00 – 18:00 (6 mins):** Direct Form II (Canonical Form). Combining delay lines, state equations, Transposed forms.
- **18:00 – 25:00 (7 mins):** Cascade (Series) Form. Factoring into second-order sections (biquads).
- **25:00 – 32:00 (7 mins):** Parallel Form. Partial fraction expansion.
- **32:00 – 35:00 (3 mins):** Coefficient Sensitivity, Quantization Effects, Lattice-Ladder structures.
- **35:00 – 40:00 (5 mins):** Checkpoints & Numerical Examples.

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2 General IIR Transfer Function

An Infinite Impulse Response (IIR) filter has a rational system function of the form:

$$H(z) = \frac{B(z)}{A(z)} = \frac{\sum_{k=0}^{M} b_k z^{-k}}{1 + \sum_{k=1}^{N} a_k z^{-k}} \tag{1}$$

By dividing the output $Y(z)$ by the input $X(z)$:

$$\frac{Y(z)}{X(z)} = \frac{\sum_{k=0}^{M} b_k z^{-k}}{1 + \sum_{k=1}^{N} a_k z^{-k}} \tag{2}$$

Cross-multiplying gives:

$$Y(z)\left(1 + \sum_{k=1}^{N} a_k z^{-k}\right) = X(z)\left(\sum_{k=0}^{M} b_k z^{-k}\right) \tag{3}$$

$$Y(z) + \sum_{k=1}^{N} a_k z^{-k} Y(z) = \sum_{k=0}^{M} b_k z^{-k} X(z) \tag{4}$$

Isolating $Y(z)$:

$$Y(z) = \sum_{k=0}^{M} b_k z^{-k} X(z) - \sum_{k=1}^{N} a_k z^{-k} Y(z) \tag{5}$$

Taking the inverse Z-transform yields the difference equation:

$$y[n] = \sum_{k=0}^{M} b_k x[n-k] - \sum_{k=1}^{N} a_k y[n-k] \tag{6}$$

Physical/Engineering Intuition: The b_k coefficients represent feedforward paths (zeros), and the a_k coefficients represent feedback paths (poles).

3 Direct Form I Structure

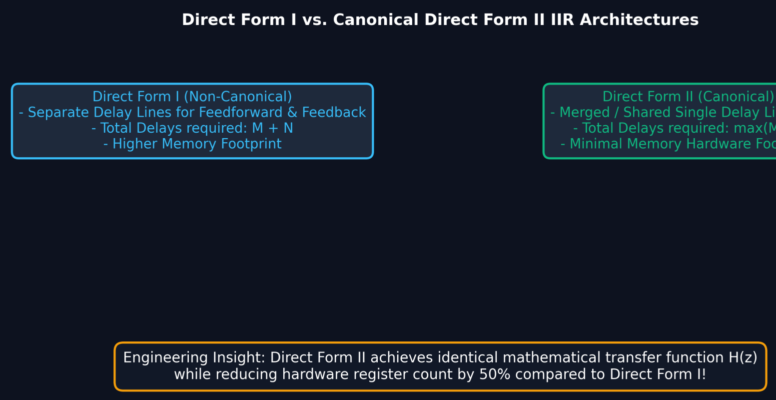

Figure 1: Direct Form I vs. Canonical Direct Form II: Achieving 50% Delay Register Savings.

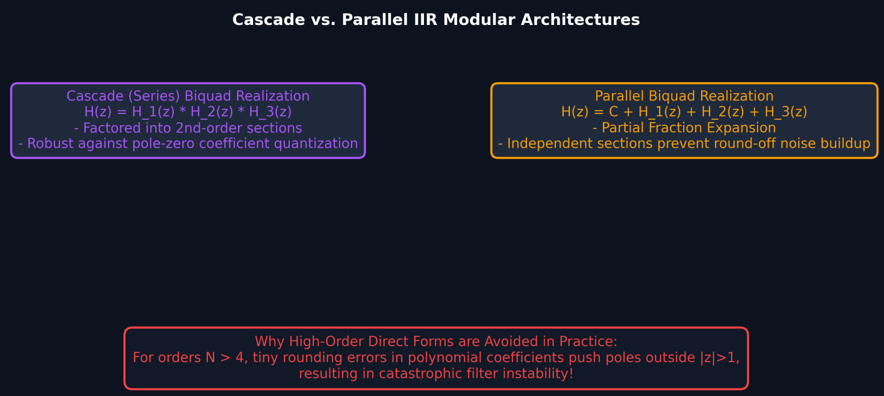

Figure 2: Cascade (Biquad Series) vs. Parallel Modular Architectures for Robust Fixed-Point Stability.

The Direct Form I structure implements the difference equation exactly as written. Let $w[n]$ be an intermediate sequence:

$$w[n] = \sum_{k=0}^{M} b_k x[n-k] \tag{7}$$

Then the output $y[n]$ is:

$$y[n] = w[n] - \sum_{k=1}^{N} a_k y[n-k] \tag{8}$$

Memory Requirements: Total memory elements $= M + N$.

4 Direct Form II (Canonical Form)

We can swap the FIR and IIR parts because they are LTI:

$$H(z) = H_{IIR}(z) \cdot H_{FIR}(z) = \left(\frac{1}{1 + \sum_{k=1}^{N} a_k z^{-k}} \right) \cdot \left(\sum_{k=0}^{M} b_k z^{-k} \right) \tag{9}$$

Let $V(z)$ be the output of the all-pole part:

$$V(z) = X(z) - \sum_{k=1}^{N} a_k z^{-k} V(z) \implies v[n] = x[n] - \sum_{k=1}^{N} a_k v[n-k] \tag{10}$$

Now, pass $V(z)$ through the FIR part:

$$Y(z) = V(z) \cdot \left(\sum_{k=0}^{M} b_k z^{-k} \right) \implies y[n] = \sum_{k=0}^{M} b_k v[n-k] \tag{11}$$

Memory Requirements: Total memory elements $= \max(M, N)$. This minimum memory form is the **Canonical Form**.

5 Cascade (Series) Form

To mitigate quantization errors, we factor $H(z)$ into 1st and 2nd-order sections (biquads).

$$H(z) = G \prod_{k=1}^{K} H_k(z) \quad \text{where} \quad H_k(z) = \frac{b_{k0} + b_{k1} z^{-1} + b_{k2} z^{-2}}{1 + a_{k1} z^{-1} + a_{k2} z^{-2}} \tag{12}$$

Complex conjugate poles and zeros form real-coefficient biquads.

6 Parallel Form

Another decomposition uses Partial Fraction Expansion:

$$H(z) = C + \sum_{k=1}^{K} H_k(z) \quad \text{where} \quad H_k(z) = \frac{\gamma_{k0} + \gamma_{k1} z^{-1}}{1 + a_{k1} z^{-1} + a_{k2} z^{-2}} \tag{13}$$

7 Coefficient Sensitivity & Quantization Effects

High-order Direct Form I suffers from extreme root sensitivity to coefficient rounding. Cascade and parallel forms localize the errors to individual 2nd-order sections, maintaining filter stability.

8 Lattice-Ladder structures

Lattice-Ladder filters use reflection coefficients (PARCOR). An all-pole (AR) filter is strictly stable if and only if $|K_m| < 1$.

9 Key Formulas

Concept	**Equation / Transfer Function**
Difference Equation	$y[n] = \sum_{k=0}^{M} b_k x[n-k] - \sum_{k=1}^{N} a_k y[n-k]$
Direct Form II State Eq	$v[n] = x[n] - \sum_{k=1}^{N} a_k v[n-k]$; $y[n] = \sum_{k=0}^{M} b_k v[n-k]$
Biquad Section (Cascade)	$H_k(z) = \frac{b_{k0}+b_{k1}z^{-1}+b_{k2}z^{-2}}{1+a_{k1}z^{-1}+a_{k2}z^{-2}}$
Parallel Section	$H_k(z) = \frac{\gamma_{k0}+\gamma_{k1}z^{-1}}{1+a_{k1}z^{-1}+a_{k2}z^{-2}}$
Stability via Lattice	$\lvert K_m\rvert < 1 \quad \forall m$

10 Checkpoint Questions

- **Q1:** An IIR filter has $H(z) = \frac{1+2z^{-1}+z^{-2}}{1-0.5z^{-1}+0.25z^{-2}}$. How many delay elements are required for Direct Form I vs Direct Form II?
  Answer: Direct Form I: $M + N = 2 + 2 = 4$. Direct Form II: $\max(M, N) = 2$.
- **Q2:** Decompose $H(z) = \frac{1}{(1-0.5z^{-1})(1-0.25z^{-1})}$ into a Parallel Form.
  Answer: Using PFE, $A = \frac{1}{1-0.25(2)} = 2$, $B = \frac{1}{1-0.5(4)} = -1$. $H(z) = \frac{2}{1-0.5z^{-1}} - \frac{1}{1-0.25z^{-1}}$.
- **Q3:** Why are Cascade/Parallel forms preferred over Direct Form I?
  Answer: High-order polynomial roots are hypersensitive to coefficient quantization, causing instability. Lower-order forms localize errors.